Lecture Scribe Generation for Joint Random Variables

L15_L17
PDF

Example Le...Scribe (1)
PDF

Using the scribe structure pdf provided in the attached "Example Lecture Scribe" PDF, generate a lecture scribe for the provided Lecture 16 (mention lecture number) materials. the scribe must have all the topics included and it's content based on example pdf. This scribe is intended to serve as a reference material for a closed-notes exam preparation. You must prioritize correctness and maintain strict faithfulness to the provided context, including definitions, notation, theorem statements, and step-by-step proofs (if any), worked examples exactly as they appear in the slides. Identify every major topic in the lecture, For EACH identified topic, you must independently repeat the given structure in example scribe format. * Do not include any creative explanations, external intuition, or material not traceable to the attached files*. Organize the output to follow the logical flow and scope of the lecture to ensure it is suitable for reliable revision.

CSE 400: Joint Random Variables

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Topic 1: Introduction to Joint Random Variables

1. Topic Title

Why Study Joint Random Variables?

2. Definitions and Notation

- **Joint Distribution**: A distribution that allows for the computation of probabilities for events involving two or more variables simultaneously and provides an understanding of their relationship.
- **Independence**: The simplest state of joint variables where they do not influence each other.
- **Dependence Measures**:
    - **Covariance**: A measure of the nature of dependence between non-independent variables.
    - **Correlation**: A measure used to understand the relationship between non-independent variables.

3. Assumptions / Conditions

- In real-life and scientific applications, we are often interested in two or more random variables (RVs) at the same time.
- Relationships are expected between simultaneous variables, such as vehicle speed and traffic density.

4. Main Results / Theorems

- **Correlation**: Variables X (Signal-to-Noise Ratio) and Y (Data Throughput) are correlated because throughput depends on SNR.
- **Application Areas**: Probabilistic analysis of joint variables is used in:
    - **Smart Transportation**: Adaptive traffic signals, congestion prediction, and autonomous vehicle routing.
    - **Wireless Networks**: 5G scheduling, adaptive modulation, and network planning.

- **Other Fields**: Weather forecasting, logistics (UAV systems), and social science.

6. Worked Examples

- **Example 1: Smart Transportation**: X = Vehicle Speed, Y = Traffic Density.
- **Example 2: Wireless Network Performance**: X = Signal-to-Noise Ratio (SNR), Y = Data Throughput.
  - **Throughput vs. SNR Table**:

SNR Level	Throughput High	Throughput Low
High	0.40	0.10
Medium	0.20	0.15
Low	0.05	0.10

- **Example 3: Weather Prediction**: X = Rainfall, Y = Temperature.
- **Example 4: Drone Delivery**: X = Wind Speed, Y = Delivery Time.

Topic 2: Discrete Joint Random Variables

1. Topic Title

Discrete Case - Joint Probability Mass Function (PMF)

2. Definitions and Notation

- **Setup**: $X \in \{x_1, x_2, \ldots, x_n\}$ and $Y \in \{y_1, y_2, \ldots, y_m\}$.
- **Product Set**: The ordered pair (X, Y) takes values in the set $\{(x_1, y_1), (x_1, y_2), \ldots, (x_n, y_m)\}$.
- **Joint PMF**: $p(x_i, y_j) = P(X = x_i, Y = y_j)$.
  - This represents the probability of joint outcomes occurring simultaneously.

3. Assumptions / Conditions

- **Non-negativity**: $0 \leq p(x_i, y_j) \leq 1$.
- **Normalization**: $\sum_{i=1}^{n} \sum_{j=1}^{m} p(x_i, y_j) = 1$.

4. Main Results / Theorems

- **Table Representation**: The joint PMF is organized in a table where rows represent X values ($x_1 \dots x_n$) and columns represent Y values ($y_1 \dots y_m$).

6. Worked Examples

- **Example: Rolling Two Dice**: Roll two fair dice. X = value on the first die; T = total sum of both dice.
  - Each outcome (i, j) has probability $1/36$.
  - **Joint Probability Table (X, T) snippet**:
    - $P(X = 1, T = 2) = 1/36$
    - $P(X = 1, T = 8) = 0$
    - $P(X = 2, T = 3) = 1/36$
  - **Observations**: Entries are either 0 or $1/36$. X and T are not independent.

Topic 3: Continuous Joint Random Variables

1. Topic Title

Continuous Case - Joint Probability Density Function (PDF)

2. Definitions and Notation

- **Joint PDF**: A function $f(x, y)$ such that $P((X, Y) \text{ lies in a small rectangle of area } dx\,dy) \approx f(x, y)dx\,dy$.
- **Product Region**: If $X \in [a, b]$ and $Y \in [c, d]$, the pair (X, Y) takes values in $[a, b] \times [c, d]$.

3. Assumptions / Conditions

- **Non-negativity**: $f(x, y) \geq 0$.
- **Normalization**: $\int_c^d \int_a^b f(x, y)dx\,dy = 1$.

- **Note**: $f(x, y)$ is a density, not a probability, and can take values greater than 1.

4. Main Results / Theorems

- **Analogous to Discrete**:
  - Discrete values $\longrightarrow$ Continuous intervals.
  - Joint PMF $\longrightarrow$ Joint PDF.
  - Sums $\longrightarrow$ Double integrals.

6. Worked Examples

- **Example**: Given $f(x, y) = 4xy$ for $0 \le x \le 1$ and $0 \le y \le 1$.
  - **Step 1: Check Validity (Normalization)**:
    $\int_0^1 \int_0^1 4xy\,dx\,dy = \int_0^1 (\int_0^1 4xy\,dx)dy = \int_0^1 2y\,dy = [y^2]_0^1 = 1.$
  - **Step 2: Compute $P(A)$ for $A = \{X < 0.5, Y > 0.5\}$**:
    $P(A) = \int_0^{0.5} \int_{0.5}^1 4xy\,dy\,dx = \int_0^{0.5} [2xy^2]_{0.5}^1 dx = \int_0^{0.5} 2x(1 - 0.25)dx = \int_0^{0.5} \frac{3}{2}x\,dx = \frac{3}{16}.$

Topic 4: Joint Cumulative Distribution Function

1. Topic Title

Joint Cumulative Distribution Function (Joint CDF)

2. Definitions and Notation

- **Notation**: $\{X \le x \text{ and } Y \le y\}$.
- **Definition**: $F(x, y) = P(X \le x, Y \le y)$.

4. Main Results / Theorems

- **Continuous Case**: $F(x, y) = \int_c^y \int_a^x f(u, v)du\,dv$.
- **Recovering PDF**: $f(x, y) = \frac{\partial^2}{\partial x \partial y} F(x, y)$.
- **Discrete Case**: $F(x, y) = \sum_{x_i \le x} \sum_{y_j \le y} p(x_i, y_j)$.

    - **Interpretation**: Add all probabilities in the table for rows where $x_i \le x$ and columns where $y_j \le y$.

5. Properties of the Joint CDF

- **Non-decreasing**: If x or y increase, $F(x, y)$ must stay constant or increase.
- **Lower-left Limit**: $\lim_{(x,y)\to(-\infty,-\infty)} F(x, y) = 0$.
- **Upper-right Limit**: $\lim_{(x,y)\to(\infty,\infty)} F(x, y) = 1$.

Would you like me to generate a set of practice problems based on these Joint Distribution examples to help you prepare for your closed-notes exam?